Measuring Detection Latency in Video World Models: A Pre-Registered Protocol and Its Failure Modes

DRAFT — results pending registered run

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Results pending. The registered experiment has not been run: it requires GPU credit that was not available, not any missing code. The harness is complete, tested and launch-ready at an estimated one GPU-hour. Every place a registered number would appear is marked **[pending registered run]** and no placeholder, estimated or illustrative value appears anywhere in this manuscript. This version contributes the protocol, the benchmark and the failure modes we found while building them.

DRAFT — results pending

Abstract

A video world model that is about to be surprised might reveal it in the *geometry* of its internal states before its prediction error rises. The claim is worth testing: prediction error is the established signal by which these models detect physical violations [Garrido et al., 2025], and representation geometry is known to carry physical plausibility on its own [Internò et al., 2026]. What is not established is which of the two crosses an alarm threshold sooner — and that comparison turns out to be unusually easy to decide in advance by accident.

We present a pre-registered, leakage-free protocol for measuring *detection latency* in frozen video world models, a synthetic benchmark of four anomaly families injected at a known frame, and a model-agnostic adapter contract that a new architecture joins in roughly twenty-five lines and is then held to by a parameterised test suite. Detection latency is scored as a survival quantity, with thresholds calibrated on held-out controls so every monitor is matched on false alarms.

Our contribution here is the apparatus and five concrete failure modes we hit while building it: a prediction-error baseline whose context has already attended to the frames it predicts, making it unfairly strong; a registered primary metric that is mathematically unable to fire under heavy censoring; a baseline whose dynamic range is small enough that it may alarm at chance, which would make beating it vacuous; an occlusion condition that measures object permanence rather than partial corruption; and a second architecture that leaks by an entirely different route. Each is either made structurally impossible, tested adversarially, or surfaced as a reported diagnostic. The registered result itself is **[pending registered run]**.

1 Introduction

A world model deployed online has to notice when the world stops behaving as it expects. The usual signal is the model's own prediction error, and for video world models this is not folklore:

Garrido et al. [2025] show that V-JEPA detects violations of intuitive physics precisely by comparing the future against its own latent prediction of it, with ablations pointing to that mechanism rather than to data or scale.

Separately, Internò et al. [2026] show that physical plausibility is implicitly encoded in the *geometry* of frozen encoder embeddings. So two signals are available from the same frozen model, and both are known to carry physical information. The open question is temporal: when a disturbance arrives, which one crosses an alarm threshold first? If the geometry moves sooner, a monitor reading it buys reaction time at no extra model cost, which is the currency that matters for a controller that must stop or re-plan.

Terminology. We say **detection latency** throughout, and avoid the phrase "early warning". Our benchmark injects each anomaly abruptly at a known frame, so what is measured is how quickly a monitor reacts to an event that has already occurred — not whether it anticipates one. The distinction matters because anticipation is a different and stronger claim, which this design cannot support.

The comparison is easy to decide in advance by accident. Four failure modes sit between a plausible experiment and a trustworthy one, and each quietly favours a predetermined answer.

1. **The baseline can cheat.** V-JEPA 2's reference implementation runs one encoder over the whole clip and then uses token masks to split context from target, so the context representations have already attended to the frames being predicted. The resulting loss is partly self-fulfilling, which makes the baseline look *stronger* than it is — biasing against the hypothesis under test. We take context from a separate encoder pass whose final tubelet is overwritten, and verify by assertion that changing the predicted frames leaves the prediction bit-identical (Section 3).

2. **Thresholds can leak.** An alarm threshold fitted on the same clips used to measure false alarms reports an optimistic operating point. Control clips are split into disjoint calibration and evaluation sets, and the learned baseline is cross-fitted within calibration so even its own threshold is honest.

3. **The registered metric can be degenerate.** Detection latency is right-censored: clips that never alarm have no observed value. Once fewer than half the clips alarm, the median is pinned at the censoring value *for every monitor*, and the comparison returns "no difference" regardless of the truth. We kept the median as the registered primary because it was specified in advance, and registered a censoring-robust secondary alongside it before running anything, so a degenerate primary is visible rather than silently null.

4. **The verdict can be vacuous.** If the prediction-error baseline does not detect the anomalies at all — if its detection rate equals its own false-alarm rate — then "geometry is faster" is true in the way that anything is faster than never. This is not hypothetical: Pavuluri et al. [2026] report exactly this collapse for a V-JEPA-based novelty signal, which looked effective across datasets but fell to chance within one. We therefore report a *lift* for every monitor, flag any baseline firing at chance, and mark an otherwise-passing verdict VACUOUS. To our knowledge this check is not standard practice in representation-based anomaly detection, though the underlying phenomenon is documented.

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What this version contributes. The registered experiment is **[pending registered run]**: it needs roughly one GPU-hour of credit that was not available, not any missing code. What we can report now is the apparatus and what building it taught us.

- A pre-registered protocol for detection latency in frozen video world models, with causality enforced structurally — monitors are streaming objects that cannot reference the future — and verified adversarially by inverting every frame after t and asserting every score is unchanged.
- A benchmark of four anomaly families injected at a fixed frame into a controlled physical scene, with matched pre-injection histories so comparisons are paired at the clip level. We characterise what each family actually measures, including an occlusion condition that turns out to probe object permanence rather than partial corruption (Section 5).
- An evaluation that treats detection latency as survival analysis — Kaplan–Meier curves, log-rank, paired Wilcoxon, and a restricted mean with a paired cluster bootstrap — under two threshold regimes, with multiplicity control across all comparisons.
- A model-agnostic adapter contract, demonstrated on two architectures from opposite sides of the pixel/latent divide, with a parameterised test suite that every registered adapter must pass. Implementing the second one showed that the leakage trap is architecture-specific: VideoMAE leaks by a completely different route (Section 7).
- Five documented failure modes, each made structurally impossible, tested adversarially, or surfaced as a reported diagnostic.

On reporting a null result. The protocol commits in advance to a decision gate and to writing up a null result if no probe clears it. We regard that commitment as the main reason to trust whatever the gate eventually returns, and the harness is built so that a null answer is as publishable, and as informative, as a positive one.

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2 Related work

Predictive world models. Learned models that roll the world forward in a latent space underpin model-based control [Ha and Schmidhuber, 2018, Hafner et al., 2020], and the joint-embedding predictive architecture (JEPA) family makes the prediction explicitly representational rather than pixel-level [LeCun, 2022, Assran et al., 2023]. V-JEPA extends this to video [Bardes et al., 2024] and V-JEPA 2 scales it [Assran et al., 2025]. We treat V-JEPA 2 as frozen throughout: the question is what its representations already encode, not what they could be trained to encode.

Prediction error as the established physics signal. Garrido et al. [2025] show that V-JEPA acquires intuitive physics from self-supervised pretraining alone, and crucially that it detects physical violations *by comparing the future against its own prediction of it* — latent prediction error is the mechanism, and ablations point to it rather than to data or scale. That result is the reason our $\mathcal{L}_0$ baseline is the model's own pretraining objective rather than a proxy: the prediction-error signal is not a straw man, it is the published mechanism, and a protocol that weakened it would be testing nothing.

Geometry of frozen embeddings already carries physical information. Internò et al. [2026] show that physical plausibility is implicitly captured by geometric properties of per-frame embeddings from frozen image encoders, with strong violation-detection results and correspondence to human responses to object-permanence violations. This matters for how our contribution should be read. That representation geometry encodes physical plausibility is established; we do not claim it. The open question we address is *temporal*: given that both the geometry and the prediction error carry the signal, which one crosses an alarm threshold sooner, measured under a protocol where neither side has been quietly advantaged. Relatedly, Kriegeskorte et al. [2008] established comparing representations by their geometry rather than their values, which motivates our probe B.

Occlusion, and what our occlusion condition actually tests. Xie and Alanwar [2026] probe exactly the regime our occlusion family lands in: a frozen V-JEPA 2 predictor forgets moving hidden objects within about 0.3 seconds, while the encoder retains hidden-object information for far longer. Our occlusion condition hides the disc completely in 77.5% of post-injection frames (Section 5), so it is an object-permanence probe, not a partial-corruption probe — and their encoder/predictor asymmetry predicts that a monitor reading encoder geometry and a monitor reading the predictor's error need not behave alike on it. We flag this as a confound to interpret against rather than a prediction we tested.

Self-supervised novelty signals can collapse to chance. Pavuluri et al. [2026] evaluate a V-JEPA-based scheme that flags clips with unpredictable embeddings as interesting. It appeared to work across datasets, but the effect was domain shift: within a single dataset performance fell to chance, while a lightly supervised probe on the same frozen features roughly doubled average precision. Their diagnosis — the representation was adequate, the self-supervised objective was the limitation — is the direct precedent for the guard we describe in Section 4. A monitor whose detection rate equals its own false-alarm rate is firing at chance, and a probe that "beats" such a baseline has demonstrated nothing. To our knowledge this check is not standard in representation-based anomaly detection; our contribution is to make it a reported quantity and to mark an otherwise-passing verdict vacuous when it fails, not to have noticed the phenomenon first.

Monitoring latents for failures, and calibrating the threshold. Zhang et al. [2026] detect failures in long-horizon manipulation from action-conditioned world-model latents using only episode-level labels, and calibrate detection thresholds with conformal prediction. Samad [2026] work in the complementary direction, showing that *anticipated* V-JEPA 2 latents are more semantically separable than observed ones and reporting an anticipatory advantage at half a second. Both monitor world-model latents, as we do; both differ from this work in that neither isolates the latent signal against the model's own prediction error under a frozen protocol with matched false-alarm rates, which is the specific comparison we are trying to make trustworthy.

Compression of world models. Fu et al. [2026] study post-training quantization of DINO-WM and report asymmetric sensitivity across components, plus failure modes that further optimization does not repair. That asymmetry is the motivation for the compression study we have drafted but deliberately not run (Section 9): if encoder and predictor degrade differently,

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a geometry monitor and a prediction-error monitor should degrade differently too, and a per-anomaly latency measurement would show it where an aggregate score would not.

Detection latency as a survival quantity. How quickly a monitor fires is a time-to-event quantity, and some episodes are never detected, so their latency is right-censored and a plain mean or median over observed alarms is biased. We use Kaplan–Meier estimation [Kaplan and Meier, 1958] and the log-rank test [Mantel, 1966], and — because a median is degenerate under heavy censoring — a restricted mean detection latency, the analogue of restricted mean survival time [Royston and Parmar, 2013]. Paired comparisons use the Wilcoxon signed-rank test [Wilcoxon, 1945] and a cluster bootstrap [Efron, 1979], with Holm's correction [Holm, 1979]. Survival analysis is uncommon in anomaly-detector evaluation, which more often reports AUROC at a fixed horizon; we argue it is the right frame whenever the quantity of interest is *how quickly*, not merely *whether*.

Pre-registration in machine learning. Freezing hypotheses, metrics and decision rules before observing data is routine in clinical trials and rare here, where the same data typically serves exploration and confirmation. We adopt it because the hypothesis is attractive, the design space is large, and the number of defensible-looking analysis choices is more than sufficient to manufacture a positive result by accident. Every departure from the frozen protocol is logged with its date and provenance (Appendix A).

3 Monitors

Let a clip be a frame sequence $f_0, \ldots, f_{T-1}$ with $T = 30$. Every monitor emits one scalar per frame and must be a function of $f_{\le t}$ alone. We enforce this structurally: each monitor is a streaming object that receives one latent at a time and holds no forward state, so referencing a future frame is not something the implementation can express. Section 4 describes the adversarial test that confirms it.

3.1 Latent state

V-JEPA 2 embeds video with a 3D convolution of stride (τ, p, p) for tubelet size $\tau = 2$ and patch size $p = 16$, followed by a flatten-and-transpose. The token sequence is therefore *t-major*: token i belongs to tubelet $\lfloor i/(H'W') \rfloor$, where $H'W'$ is the number of spatial patches. Tubelet j owns the contiguous index block $[\, jH'W',\ (j{+}1)H'W' \,)$. Every index computation below rests on this, and we verify it directly against the patch embedder rather than assuming it.

Because the tubelet spans two frames, no V-JEPA 2 representation is a function of a single frame; the protocol's notion of a "per-frame latent" has to be made precise. Define the **causal window** ending at t,

$$W_t \;=\; \big(f_{\max(t-T_{\mathrm{ctx}}+1,\,0)},\ \ldots,\ f_t\big), \qquad T_{\mathrm{ctx}} = 16, \tag{1}$$

left-padded by repeating f_0. Let $E(\cdot)$ be the frozen encoder and $E_j(\cdot)$ its output tokens for tubelet j. With $J = T_{\mathrm{ctx}}/\tau - 1$ the final tubelet index, the latent state is the spatial mean of the final tubelet,

$$z_t \;=\; \frac{1}{H'W'} \sum_k \big[E_J(W_t)\big]_k \;\in\; \mathbb{R}^d, \qquad d = 1024. \tag{2}$$

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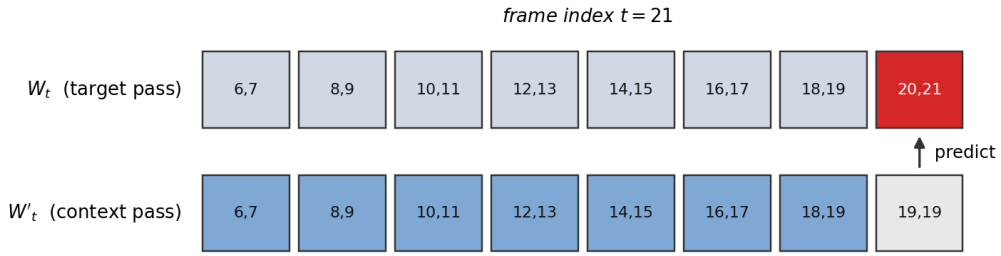


Figure 1: The two encoder passes used at frame t. Each box is one 2-frame tubelet, labelled by the frames it spans. The context pass W'_t overwrites the final tubelet with a repeat of f_{t-2}, so the predictor's context is a function of $f_{\leq t-2}$ only, while the target comes from the full window W_t. A single-pass implementation would let the context attend to f_{t-1}, f_t and make the loss partly self-fulfilling.

The final tubelet spans frames $(t-1, t)$, so z_t depends on no frame later than t. All three probes are functions of the sequence $\{z_t\}$ alone.

3.2 Baseline $\mathcal{L}_0$: the model's own loss, evaluated causally

V-JEPA 2's pretraining objective predicts the representation of masked target tokens from context tokens. Evaluating it online requires care. The reference implementation runs *one* encoder pass over the full video and then applies masks, so the context tokens have attended to the frames being predicted; its own source notes that training instead uses a separate target encoder. Using that single-pass loss as our baseline would let the context see the answer, inflating the baseline and biasing against the hypothesis.

We therefore use two encoder passes. Define the **context window**

$$W'_t \;=\; (f_{t-15},\, \ldots,\, f_{t-2},\, f_{t-2},\, f_{t-2}), \tag{3}$$

that is, W_t with its final tubelet overwritten by a repeat of f_{t-2}. Tubelets $0..J-1$ of W'_t are exactly frames $f_{t-15..t-2}$, and no token in them can attend to f_{t-1} or f_t (Figure 1). Writing $P(\cdot \mid \cdot)$ for the frozen predictor applied to those context tokens with the final tubelet's positions as targets,

$$\mathcal{L}_0(t) \;=\; \left\| \, P\big(E_{0:J-1}(W'_t)\big) \;-\; E_J(W_t) \, \right\|_2^2 / (H'W'd), \tag{4}$$

the mean squared error between the prediction and the true final-tubelet tokens. The target is allowed to depend on f_t — it *is* the quantity revealed at time t — while the prediction is not.

3.3 Baseline $\mathcal{L}_1$: information-matched prediction error

$\mathcal{L}_0$ and the geometry probes consume different representations — token grids versus a pooled vector — which invites the objection that any difference between them is an artefact of what they see. We therefore add a second baseline that consumes *exactly* the sequence the probes do. Let $W^\star$ be a ridge map fitted to predict z_t from its two predecessors,

$$W^\star = \arg\min_W \sum \| z_t - W[z_{t-1}; z_{t-2}] \|_2^2 + \lambda \|W\|_F^2, \qquad \lambda = 1, \tag{5}$$

fitted on control-calibration clips only, and $\mathcal{L}_1(t) = \|z_t - W^\star[z_{t-1}; z_{t-2}]\|_2$. Scores for the calibration clips themselves are produced by 5-fold cross-fitting, so $\mathcal{L}_1$'s own threshold is not optimistic. A probe must beat *both* baselines to count.

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3.4 Geometry probes

All three are computed online from $\{z_t\}$ with no normalisation.

$$\text{A. Trajectory curvature:} \quad a_t = \left\| z_t - 2z_{t-1} + z_{t-2} \right\|_2 \tag{6}$$

$$\text{B. Representational distance:} \quad b_t = 1 - \frac{\langle z_t,\ \bar{z}_t \rangle}{\|z_t\|\,\|\bar{z}_t\|}, \qquad \bar{z}_t = \tfrac{1}{5}\textstyle\sum_{i=1}^{5} z_{t-i} \tag{7}$$

$$\text{C. Covariance shift:} \quad c_t = \left\| \Sigma_t - \Sigma_{\text{base}} \right\|_F, \qquad \Sigma_t = \widehat{\text{Cov}}(z_{t-4..t}) \tag{8}$$

where $\Sigma_{\text{base}} = \widehat{\text{Cov}}(z_{0..14})$ is frozen at $t = 14$, and both covariances use ddof $= 1$. Probe A is a second difference, probe B is scale-invariant, probe C compares second-order structure against the clip's own pre-injection baseline.

Computing probe C without forming $d \times d$ matrices. Written naively, (8) builds two 1024×1024 covariances per frame. Both are low-rank: with centred factors $A \in \mathbb{R}^{k_a \times d}$ and $B \in \mathbb{R}^{k_b \times d}$ and $\Sigma = A^\top A/(k{-}1)$, the trace identities give

$$\|\Sigma_a - \Sigma_b\|_F^2 = \frac{\|AA^\top\|_F^2}{(k_a{-}1)^2} - \frac{2\,\|AB^\top\|_F^2}{(k_a{-}1)(k_b{-}1)} + \frac{\|BB^\top\|_F^2}{(k_b{-}1)^2}, \tag{9}$$

which costs $O(k^2 d)$ instead of $O(d^2 k)$ and touches only $k \times k$ Grams. We verify it against the naive computation to machine precision ($\leq 2 \times 10^{-16}$ relative).

A numerical caveat. Probe A is a second difference of nearby vectors, the textbook setting for catastrophic cancellation: relative error in z is amplified. Reduced-precision inference is therefore not a free speed-up here, and we measure fp32 against bf16 and fp16 on the actual probe values rather than assuming a choice is safe.

4 Evaluation protocol

4.1 Thresholds, and what "fair" means here

Monitors live on incomparable scales — a Frobenius norm and a cosine distance have no common unit — so comparing their raw scores is meaningless. What can be matched is their false-alarm behaviour. For each monitor independently, the threshold is the 99th percentile of that monitor's scores pooled over the *control-calibration* clips and the evaluation frames. Every monitor therefore raises false alarms on 1% of control frames by construction, and differences in detection latency are differences in sensitivity, not in scale.

Control clips are split into disjoint calibration ($n =$ PENDING) and evaluation ($n =$ PENDING) sets. Thresholds are set on the former and false alarms measured on the latter; no threshold ever sees anomalous data.

A second operating point. Matching the *per-frame* rate at 1% compounds to roughly 14% over a 15-frame episode, which no episode-level monitor could use. We therefore repeat the entire analysis with thresholds re-derived so the *clip-level* false-alarm rate on calibration controls is 5%. This was fixed before any inference run, and both regimes are reported unconditionally, so neither can be selected after the fact.

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4.2 Detection latency as a survival quantity

An alarm fires at the first evaluation frame whose score exceeds the threshold. With injection at $t^\star = 15$ and an evaluation window of frames 15..29, detection latency is $\Delta t = t_{\text{alarm}} - 15 \in \{0, \ldots, 14\}$. A clip that never alarms is **right-censored** at 14: its latency is not 14, it is unobserved and at least 14.

This censoring is the reason the protocol's headline metric needs a companion. Once fewer than half the clips alarm, the median Δt equals the censoring value for *every* monitor, and the pre-registered comparison returns "no difference" no matter how large the true effect. We registered the median as primary because it was specified in advance, and registered the **restricted mean detection latency**

$$\text{RMDL} \;=\; \mathbb{E}\big[\min(\Delta t, 14)\big] \;=\; \sum_{u=0}^{13} \widehat{S}(u), \tag{10}$$

the area under the Kaplan–Meier curve $\widehat{S}$, as a secondary that stays well defined at any detection rate. Lower is earlier.

4.3 Tests

- **Primary (as registered).** Paired Wilcoxon signed-rank on Δt, probe versus baseline, one-sided (the hypothesis is directional), with censored clips entered at 14 — conservative, since it understates differences. The two-sided p is always reported alongside, so the tail cannot be chosen post hoc.
- **Censoring-aware.** Log-rank on the Kaplan–Meier curves. Both arms come from the same clips, violating its independence assumption, so it is reported as corroborative only.
- **Effect size.** Paired cluster bootstrap on the RMDL difference (PENDING resamples, the same resample applied to both monitors so that pairing is preserved), giving a confidence interval in frames.
- **Multiplicity.** Holm's step-down correction across all six comparisons (3 probes × 2 baselines), separately for the median and RMDL families.

4.4 The decision gate

A probe counts as a **faster detector** iff, against *both* baselines, its median Δt is at least one frame lower *and* the Holm-adjusted one-sided p is below 0.01. If no probe qualifies, the protocol terminates the expansion track and the result is written up as a negative one. The gate is evaluated by code, not by reading tables.

4.5 Pre-specified follow-up: what lives in the lag window

If a probe does alarm before the loss, the frames in between are where the representation has registered something the output error has not, and the natural question is what. We pre-specify the analysis here so that it is fixed before the lag window is known: fit linear probes to the latents z_t for $t \in [t^\star + \Delta t_{\text{probe}},\ t^\star + \Delta t_{\text{loss}})$, targeting (i) which of the four anomalies

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was injected, and (ii) the disc's position and velocity, which are the quantities the physics conditions actually change.

Two controls are part of the specification, because a linear probe on high-dimensional latents is easy to over-read. Frames from one clip are highly correlated, so a random train/test split lets the probe memorise clip identity and return a near-perfect score that means nothing; all cross-validation is therefore grouped by clip. And an accuracy is uninterpretable without a reference, so every score is reported beside the majority-class rate and a *clip-level* label-shuffled control, which must land at chance. If the shuffled control does not land at chance, the split is leaking and the real score is not reported as evidence.

This analysis is contingent on the gate and is listed here as method, not result.

4.6 Guards against a vacuous verdict

Two checks sit outside the hypothesis tests and exist to stop the gate from being believed when it should not be.

Lift. For every monitor we report lift = detection rate − held-out clip false-alarm rate. A monitor with lift near zero is firing at chance. If a *baseline* fires at chance, then beating it carries no information about H1 — a probe cannot be shown to fire *sooner* than a signal that never fires — and a passing verdict is reported as VACUOUS. This check is not a hypothesis test and does not alter any threshold or statistic; it only makes an existing failure mode visible.

Causality, tested adversarially. Claiming that frame t cannot see frame $t{+}1$ is cheap; testing it is not. For every registered adapter we take a clip, invert every frame after a cut point, and assert that z_t, $\mathcal{L}_0$ and all three probes are unchanged to floating-point tolerance for t at or before the cut. A separate test asserts that the $\mathcal{L}_0$ *prediction* does not move when only f_{t-1} and f_t change, while confirming it does move when f_{t-2} changes — so the guarantee is not vacuously satisfied by a prediction that ignores its input. Probe streams are additionally checked for prefix invariance: scores for frames $0..t$ must be identical whether or not later frames exist.

5 Benchmark

A disc of radius 5 bounces inside a 64×64 frame for 30 frames under constant downward gravity $g_0 = 0.25\,\mathrm{px/frame}^2$ with perfectly elastic collisions, so the baseline dynamics conserve energy and are statistically stationary. Initial position, speed and direction are randomised; generation is deterministic given the frozen seed.

At frame $t^\star = 15$ one of four anomalies is injected (Figure 2):

gravity inverted $g \to -g_0$. Acceleration changes immediately; position diverges only gradually, which makes this the subtlest condition at short latency.

friction per-frame velocity retention $0.8^{1/5}$ and bounce restitution 0.8. Applied per frame rather than only at collisions, so the effect is observable from the first post-injection frame; a collision-only version would leave detection latency undefined on clips with no bounce in the window.

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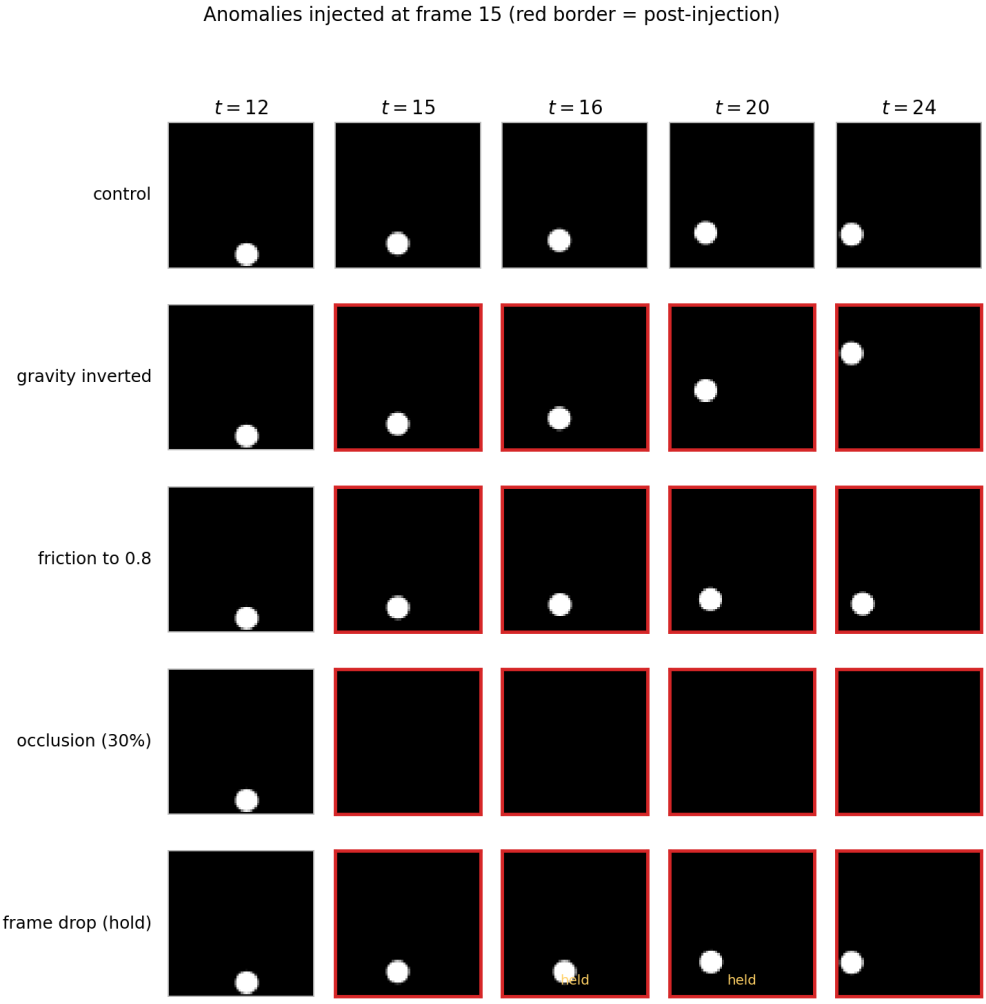


Figure 2: The four anomaly families, one row each, against the control row. Red borders mark post-injection frames; "held" marks a frame the sensor glitch repeats. All conditions share the control's pre-injection history exactly, so every comparison is paired at the clip level.

occlusion a static black square of 29.9% frame area, anchored on the disc's path.

frame drop for underlying frames 15..22, every other frame is not delivered and the previous frame is held in its place — four held frames — with normal delivery resuming at frame 23.

What the occlusion condition actually tests. A 35×35 square anchored on the trajectory of a 10-pixel disc does not "partially block" it. Measured across the condition, 77.5% of post-injection frames hide the disc completely, 17.2% hide it partially, and in 28.0% of clips the disc is never visible again inside the window. This condition therefore probes object permanence — does the representation register that a tracked object has vanished — rather than robustness to partial visual corruption. We report it as such. It follows directly from the registered specification and is not a defect, but describing it as partial occlusion would misstate what the

numbers mean.

Effect sizes are not comparable across conditions. Gravity inversion changes acceleration at once; friction changes velocity by about 4.4% per frame; occlusion removes the object; the frame drop freezes it. These are different magnitudes of disturbance, so pooled results average over heterogeneous effects, and per-condition results are reported separately.

Paired design. An anomalous clip and its control share a random seed, so their frames are identical up to $t^\star$ and diverge only because of the injection. Every monitor comparison is therefore within-clip, which is what the paired tests in Section 4 exploit.

Scale. The registered design is 1000 anomalous clips (250 per condition) plus 200 controls split into PENDING calibration and PENDING evaluation. Clips are bilinearly upsampled to the model's native resolution and replicated to three channels.

6 Results

6.1 Registered experiment: [pending registered run]

The registered Phase 2 experiment — 1000 anomalous and 200 control clips at 256 px through frozen V-JEPA 2, under the protocol of Sections 3 and 4 — has **not been run**. It requires approximately one GPU-hour of credit that was not available; the harness is complete, tested, and launch-ready, and the run is recorded with its exact launch command in the repository's parked-experiment register.

Consequently this section reports **no registered results**. We have deliberately not filled the gap with illustrative, estimated or placeholder numbers. The manuscript is generated from run outputs by a script (Section 7); with no run analysed, that script emits visible PENDING markers rather than plausible values, and the following quantities are all **[pending registered run]**:

- detection latency per monitor, under both threshold regimes;
- the paired probe-versus-baseline contrasts and their corrected p-values;
- Kaplan–Meier curves and the restricted mean detection latency;
- per-anomaly-family breakdowns;
- the lift of each monitor, and therefore whether either baseline fires at chance;
- the Phase 3 gate verdict.

We state plainly that the central empirical question of this work — whether representation geometry crosses an alarm threshold sooner than the model's own prediction error — **remains open**. Nothing below should be read as bearing on it.

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6.2 Exploratory CPU pilot (not the registered experiment)

To size the registered run and sanity-check the pipeline against real pretrained weights, we began a reduced-resolution pilot on CPU. It is **exploratory**, it is **not** the registered experiment, and it cannot substitute for one.

- **Resolution.** 128 px rather than the registered 256 px, which shrinks the patch grid from 16×16 to 8×8. Context length, tubelet size, all probe definitions, the $\mathcal{L}_0$ construction, thresholds and metrics are unchanged. We rejected 96 px because the disc would then be smaller than a single patch.
- **Scale.** A reduced design of 280 clips rather than 1200.
- **Status at the time of writing:** 138 of 280 clips extracted — 70 calibration controls, 0 evaluation controls, 68 anomalous. The run is in progress; control clips incomplete, so no thresholds can be calibrated and no result can yet be computed from it.

No analysis of the pilot is reported here. Thresholds are calibrated on control clips, so until the control splits are complete no threshold exists, no alarm time is defined, and no detection latency can be computed. Reporting partial anomaly data without calibrated thresholds would be meaningless rather than merely preliminary. The pilot ordering was arranged so that controls are extracted first, precisely so that an interrupted run becomes a smaller but coherent experiment rather than an incoherent one; at the time of writing that threshold has not yet been crossed.

Two observations from a two-clip correctness check on the real pretrained weights are reported in Section 8 as risks to the registered run's power. They concern the *magnitude and dispersion* of monitor scores, not detection latency, and they are not results about the hypothesis.

6.3 Numerical precision is not neutral between the two monitor families

This is a measurement on the real pretrained weights, and it is reported because it is a real result — but it says nothing about the hypothesis, so it is not a registered outcome.

We compared fp32, bf16 and fp16 on the actual monitor values (Table 1). Reduced precision does not degrade the monitors evenly. Under bf16, the worst geometry probe shifts by a median 6.74% while $\mathcal{L}_0$ shifts by 0.23% — roughly a factor of 30. The mechanism is not subtle: $\mathcal{L}_0$ is a mean over $H'W' \times d$ values, so independent rounding error averages away, whereas probe A is a second difference of nearby vectors and probe B a cosine against a mean of five nearly identical vectors. Both amplify exactly the error the loss suppresses.

The consequence matters for anyone running this comparison. Choosing a reduced precision for throughput would have handicapped the probes while leaving the baseline essentially intact — a *directional* bias against the hypothesis, introduced by what looks like an implementation detail and invisible in any aggregate accuracy check. We therefore run the registered experiment in fp32, as pre-registered, at roughly 2.9× the cost per clip.

We flag this as a general hazard: in any comparison between a pooled output error and a higher-order statistic of internal states, the two sides have different numerical conditioning, and the precision setting is part of the experimental design rather than a free parameter.

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Precision	Monitor	median rel. err.	max rel. err.	Spearman
bf16	$\mathcal{L}_0$	0.23%	1.70%	0.9823
bf16	probe A	4.55%	19.87%	0.9863
bf16	probe B	6.74%	34.05%	0.9769
bf16	probe C	1.58%	2.95%	0.9983
fp16	$\mathcal{L}_0$	0.12%	0.73%	0.9926
fp16	probe A	1.61%	10.04%	0.9967
fp16	probe B	1.48%	8.18%	0.9981
fp16	probe C	0.48%	1.56%	0.9971

Table 1: Relative error against fp32 on the real pretrained weights, measured on the monitor values themselves. Spearman is rank correlation with the fp32 ordering. The loss is robust; the geometry probes are not.

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6.4 What the harness does produce

The apparatus itself is verified, and these are results about the protocol rather than about world models:

- Every monitor is a streaming object that cannot reference a future frame. Inverting all frames after a cut point leaves z_t, $\mathcal{L}_0$ and all three probes bit-identical for t at or before the cut, including against the real V-JEPA 2 architecture.
- The $\mathcal{L}_0$ prediction does not move when only the predicted frames change, and does move when f_{t-2} changes — so the causality guarantee is not satisfied vacuously by a prediction that ignores its input.
- Probe values match their closed forms to 10^{-9}, and the $O(k^2 d)$ covariance identity of Section 3 matches the naive $d \times d$ computation to machine precision.
- Thresholds are a function of calibration controls alone.
- Both registered adapters — V-JEPA 2 and VideoMAE — pass the identical contract suite.
- The full orchestration, including fan-out, merge, analysis and the gate, was exercised end-to-end at the full 1200-clip scale using randomly initialised weights, so the only unexercised step in the registered run is the GPU forward itself.

7 The harness as the reusable artefact

The question in this paper is one instance of a template: *does some internal signal of a frozen sequence model move before its output error does?* Answering it for another architecture, another anomaly family or another probe should not require rebuilding the protocol, and if it does, the resulting numbers will not be comparable to ours.

What is shared and what is pluggable. The causal loop — which frames frame t may see, where the context window stops, which frames receive a prediction loss — is identical for every architecture and is implemented exactly once. A model supplies two methods:

```
encode_window(windows) -> (latent, loss_target)
predict_final_step(context_windows) -> prediction
```

The windows handed to both are already the causal and context windows of Section 3, so a new adapter cannot accidentally read the future: the information boundary is enforced by the caller, not by the contributor's discipline. Probes, thresholds, survival statistics, multiplicity correction and the gate are then shared verbatim. Adding V-JEPA 2 takes roughly twenty-five lines beyond preprocessing; a model with no native prediction head returns `None` and is scored against $\mathcal{L}_1$ alone.

A second architecture, as a demonstration. To check that the contract is genuinely architecture-agnostic rather than V-JEPA-shaped, we implemented a second adapter for VideoMAE. It is the informative contrast: a masked autoencoder that reconstructs *pixels*, against a joint-embedding architecture that predicts *representations*, sharing a tubelet size and patch size so the mapping is clean. It passes every contract test.

Implementing it also surfaced that the leakage trap is architecture-specific. `VideoMAEForPreTraining` constructs its regression targets by patchifying *the same tensor it is handed*. Passing it the context window would therefore score the prediction against the overwritten pixels rather than the true frames $t-1, t$, giving a near-zero loss and a baseline that looks far stronger than it is — the same failure as V-JEPA 2's single-pass loss, reached by a completely different route. The adapter predicts from the context window while patchifying the target from the full window, replicating the model's own per-patch normalisation. The general lesson is that "is this loss causal?" has to be re-asked, and re-tested, for every architecture; it is not a property the harness can confer.

Contributions are held to the contract automatically. Every registered adapter is run through a parameterised test suite: output shapes, the loss window, prefix invariance, the future-inversion causality check, and the assertion that the prediction is blind to the frames it predicts while still responding to f_{t-2}. Registering an adapter without supplying a cheap constructor for these tests fails the suite by design, so an architecture cannot silently enter the benchmark without being held to the same standard as the others. This is what licenses a cross-architecture comparison: the difference between two rows is a difference between models, not between pipelines.

Reproducibility. The dataset is generated deterministically from a frozen seed, so no data needs to be distributed and no version can drift. The frozen constants are pinned by a test that fails if any of them is edited. Each run records its own metadata, and the analysis emits the full result set — both threshold regimes, both metric families, per-condition breakdowns, false-alarm rates and the gate verdict — in a single JSON from which every table and numeric sentence in this paper is regenerated. No number here is transcribed by hand, which removes an entire class of silent error between the experiment and the manuscript.

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Scope of the claim. We make no claim that these three probes are the best possible geometric monitors, nor that this synthetic domain is representative of deployment. The reusable part is the protocol that decides such questions without accidentally answering them in advance.

8 Limitations

No registered result yet. The first and largest limitation is that the central question is unanswered. This version contributes a protocol, a benchmark and a set of documented failure modes; it does not contribute a measurement of detection latency in V-JEPA 2. Until the registered run executes, every claim here is about the apparatus.

Sudden onset measures latency, not anticipation. Anomalies are injected abruptly at a known frame, so a monitor can only react to something that has already happened. This design cannot distinguish a monitor that detects quickly from one that anticipates, and we make no anticipation claim. Work on *anticipated* latents [Samad, 2026] addresses the other regime, where the quantity of interest is a model's forward prediction rather than its reaction. A gradual-onset benchmark would be needed to speak to anticipation, and we do not have one.

Synthetic clips. A single disc on a black background at 64×64 is a deliberately controlled setting: it makes the injection frame exact, the pre-injection histories matched, and the ground truth available. It is also nothing like deployment. Scene complexity, texture, camera motion, multiple objects and natural statistics are all absent, and any result here is a statement about this regime only.

One model family. The registered experiment covers frozen V-JEPA 2. A VideoMAE adapter exists and passes the contract suite, but *has not been run* — an adapter is a capability, not a result, and we are careful not to present one as the other. Whether any effect is a property of joint-embedding predictive architectures, of video transformers generally, or of one checkpoint is not decided here.

The occlusion family measures object permanence. Our occlusion condition hides the disc completely in 77.5% of post-injection frames, so it asks whether the representation registers that a tracked object has vanished. Xie and Alanwar [2026] show that this regime dissociates sharply inside V-JEPA 2: the predictor forgets moving hidden objects within about 0.3 seconds while the encoder retains such information for far longer. Since our geometry probes read pooled encoder states and $\mathcal{L}_0$ reads the predictor, that dissociation is a plausible confound for this family specifically, and results on it should be interpreted against their finding rather than as a clean test of the hypothesis.

Mean pooling discards where the anomaly is. Both the latent state and the $\mathcal{L}_0$ loss average over the spatial token grid. Our anomalies are spatially local, so averaging dilutes them. On two clips of real pretrained weights, $\mathcal{L}_0$ spanned only about $\pm 2\%$, which is the observation behind the lift diagnostic: if that holds at scale, $\mathcal{L}_0$ may alarm at chance and comparisons against it would be vacuous. Max-pooling, or a loss restricted to tokens whose content actually changed, are natural variants we did not register.

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Probes A and C may carry per-clip offsets. In the same two-clip check, probe C differed by roughly 8% between clips while moving only a few percent *within* each. With thresholds pooled across clips, a monitor whose between-clip variance exceeds its within-clip signal behaves partly as a per-clip constant rather than a change detector. Probe B, scale-invariant by construction, showed no comparable offset. Standardising each probe against its own pre-injection frames is the standard remedy, but it cannot be applied uniformly here because probe C is undefined before frame 15 and so has no within-clip reference window. We flag this rather than patch it: changing a frozen probe definition after seeing data is exactly what pre-registration exists to prevent.

Pooled thresholds assume stationarity in t. For $t < 15$ the causal window is left-padded by repeating f_0, so warm-up latents are computed on partly synthetic windows and the score distribution is mildly t-dependent. Controls and anomalies pass through an identical pipeline, so this cannot manufacture a difference *between* monitors, but it widens the pooled calibration distribution and costs sensitivity. A per-frame threshold would be the cleaner estimator.

Latency is quantised by the tubelet. Since z_t summarises frames $(t-1, t)$, latencies should not be read as single-frame reaction times. Both monitor families inherit the same quantisation, so comparisons stay fair, but absolute numbers are coarser than the units suggest.

The log-rank test is mis-specified here. Both arms come from the same clips, violating its independence assumption. It is reported as corroborative only; the paired Wilcoxon and the paired cluster bootstrap respect the design.

A null result would be about these probes. If no probe clears the gate, the honest conclusion is that *these three second-order summaries of a mean-pooled latent* are not faster than prediction error on *this* benchmark — not that no geometric signal exists. The space of candidate probes is large; we fixed three in advance precisely so we could not search it.

9 Future work

Run the registered experiment. The immediate next step is the one this paper is missing: execute the frozen protocol at 256 px and report the verdict, whichever way it falls. The estimated cost is about one GPU-hour. Everything needed is written and tested.

Compression and detection latency (drafted, deliberately not run). The natural follow-up becomes available only once a monitoring harness exists. Fu et al. [2026] report that post-training quantization of a world model produces *asymmetric* sensitivity across components, and failure modes that further optimization does not repair. A mean-performance number hides that; a per-anomaly-family detection-latency measurement should not.

We have drafted a second study asking whether compression changes *which* anomalies a model still catches and *how fast*, rather than degrading it uniformly. Its shape:

- Six precisions (fp32, bf16, fp16, int8 weight-only, W8A8, 4-bit weight-only) crossed with three placements (encoder only, predictor only, both). Placement is the factor that tests

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the asymmetry claim directly: under encoder-only compression the latents the probes consume are degraded while the prediction head is intact, and vice versa.

- One primary contrast per anomaly family, Holm-corrected across the four, with every other cell exploratory and labelled as such.
- **Equivalence testing**, not null-hypothesis testing. The interesting claim is usually that compression does *not* hurt, which is an equivalence claim and cannot be supported by failing to reject a difference. The draft specifies TOST on the latency difference — and leaves the equivalence margins blank, as an explicit TODO for the author to set and justify before freezing, because choosing a margin after seeing the effect is the easiest way to manufacture an equivalence result.
- A fresh seed for the confirmatory clips, so that no clip used to develop the harness appears in the confirmatory set, and per-cell threshold calibration, so that a shift in score scale is not confounded with a change in sensitivity.
- A second-GPU arm. Quantization kernels, tensor-core paths and accumulation order differ across hardware, so a compression effect that does not reproduce on two GPUs is a kernel artefact rather than a property of the model.
- Two exploratory analyses: *blind spots* — whether misses concentrate in one anomaly family rather than thinning uniformly — and *motion energy*, testing whether compression costs concentrate in low-motion clips where there is least signal to work with. Both target the same idea, that aggregate accuracy can hold while the distribution of failures shifts.

The draft is marked DRAFT, NOT FROZEN and no data may be collected under it until its margins are set and it is frozen in its own commit. It does not amend the v1 pre-registration, which remains in force untouched. We record it here so that its existence and date are on the record and it cannot later be presented as having been registered earlier than it was.

Other directions. A physics-simulator domain with contact and articulation would test whether anything here survives outside synthetic discs. A gradual-onset benchmark would let the anticipation question be asked properly, rather than the latency question we can currently answer. And the per-clip standardisation issue raised in Section 8 deserves a clean treatment: a probe family defined to be invariant to per-clip offsets from the start, rather than one patched after the fact.

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10 Conclusion

We set out to measure whether the internal geometry of a frozen video world model crosses an alarm threshold sooner than the model's own prediction error, and built the experiment so the answer could come out either way. That measurement is **[pending registered run]**: it needs about one GPU-hour of credit, not any missing code. We report the apparatus, and what building it revealed.

The methodological point stands independently of the eventual verdict. A comparison between an internal signal and an output error is unusually easy to settle in advance without noticing: by letting the baseline's context attend to the frames it predicts; by calibrating

thresholds on the data they are scored against; by summarising a censored latency with a statistic that is constant under censoring; or by beating a baseline that never fires. We made the first structurally impossible and tested it adversarially, the second impossible by construction, the third visible through a registered secondary, and the fourth visible through a lift check that marks an otherwise-passing verdict vacuous.

Two further findings came from building rather than from measuring. Our occlusion condition, specified as a square covering 30% of the frame, in practice hides the disc entirely in 77.5% of post-injection frames — so it probes object permanence, a regime in which the encoder and predictor of V-JEPA 2 are known to dissociate [Xie and Alanwar, 2026], and describing it as partial occlusion would have misstated what the numbers mean. And implementing a second architecture showed that the leakage trap is architecture-specific: VideoMAE builds its targets by patchifying whatever tensor it is handed, reaching the same bias by an entirely different route. Whether a loss is causal has to be re-asked, and re-tested, per architecture; it is not a property a harness can confer.

None of these guards is sophisticated. What they share is that omitting any one of them yields a cleaner-looking result, which is why we think they belong in the protocol rather than in the discussion. To our knowledge the lift check in particular is not standard in representation-based anomaly detection, though the collapse it guards against is documented [Pavuluri et al., 2026].

We release the harness as a benchmark that another architecture joins in about twenty-five lines and is then held to the same contract automatically, so that the next claim of this shape can be checked rather than argued — including ours, once the run happens.

Reproducibility statement

The protocol was frozen before any code was written and is reproduced in Appendix A; every subsequent change is logged with its date and the evidence that motivated it. The dataset is generated deterministically from a single seed, so none is distributed. Frozen constants are pinned by a test that fails if edited. The causality and leakage guarantees are tested, not asserted. Every table, figure and numeric sentence in this paper is generated from a run's result JSON by a script in the repository; with no run analysed, the manuscript compiles with visible PENDING markers rather than plausible values, which is why no registered number appears above.

DRAFT — results pending

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A Frozen protocol and change log

A.1 What was frozen, and when

The protocol was frozen on 2026-10-06 before any code was written. It fixes: the dataset (resolution, length, injection frame, the four anomaly definitions, the control split), the five monitors and their closed forms, the threshold rule, the detection latency metric, the statistical tests, the multiplicity correction, and the decision gate with its termination branch. The full text ships with the repository.

A.2 Contradictions in the source specification, resolved before freezing

Three requirements could not all hold simultaneously. Each was resolved before any code ran and the resolution was itself frozen.

Constant velocity versus inverted gravity. A constant-velocity baseline has no gravity to invert. Resolved by giving the baseline a constant downward g_0 with perfectly elastic collisions, which keeps the dynamics stationary and makes $g \to -g_0$ well defined.

Friction with no observable effect. A restitution change applied only at collisions may not manifest before the evaluation window closes, leaving detection latency undefined on a substantial fraction of clips. Resolved by applying a per-frame drag in addition to the restitution change.

A per-frame latent under a two-frame tubelet. V-JEPA 2 has no representation that is a function of a single frame. Resolved by the sliding causal window of Section 3, which is a function of $f_{\leq t}$ and whose final tubelet spans $(t-1, t)$.

A.3 Changes after freezing

Before any inference run.

1. The Wilcoxon tail was fixed as one-sided primary, with the two-sided value always reported. The hypothesis is directional, and the tail had to be fixed before seeing data.

2. Two secondary analyses were registered: the restricted mean detection latency with a paired cluster bootstrap, and a clip-matched 5% false-alarm-rate threshold regime. Both weaknesses follow from the *definition* of the primary metric rather than from any observed result — a censored median is degenerate below 50% detection, and 1% per frame compounds to roughly 14% per clip over fifteen frames. The primary metric and the gate were not changed.

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3. $\mathcal{L}_1$'s scores on calibration clips were specified to come from 5-fold cross-fitting, so that its own threshold is not optimistic.

After a two-clip correctness check on real weights. The lift column, the at-chance flag and the VACUOUS marker were added. What had been observed at that point was score *magnitude* — that $\mathcal{L}_0$ spanned about $\pm 2\%$ — and not any detection latency, gate outcome or effect direction. No threshold, metric, test or gate criterion was altered; the addition only makes an existing failure mode visible in the report. We record the provenance here rather than omitting it, so that a reader can judge the timing for themselves.

A.4 Parking, 2026-10-10

The registered experiment is **parked, not amended or cancelled**. No data has been collected under the frozen protocol, which remains in force and untouched. The reason is access to GPU credit, not anything about the science or the code: the run is launch-ready and estimated at about one GPU-hour. It is recorded, with its exact launch command, in the repository's parked-experiment register.

A second study on compression and detection latency has been *drafted* (Section 9). That draft is explicitly not frozen, its equivalence margins are left blank as a TODO, and no data may be collected under it. We record its existence and date here so that it cannot later be presented as having been registered earlier than it was.

The manuscript uses "detection latency" where the frozen document names its gate criterion "Early Warner". Only the wording in the paper differs; the criterion itself is unchanged. We avoid the original phrase because the benchmark injects anomalies abruptly, so the experiment measures reaction time and not anticipation.

A.5 Falsification

The hypothesis is falsified if every geometry probe has a median Δt greater than or equal to that of $\mathcal{L}_0$, or if no difference reaches Holm-adjusted $p < 0.01$. The protocol commits in that case to terminating the expansion track and reporting the negative result.

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